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Data Warehousing

Lab 2 Report

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This report outlines the five stages involved in creating a Sales Data Mart from the existing AdventureWorks2014 database. The final stage highlights an analytical preview of the sales data, presented using the Power BI tool.

Stage 1: Creating Database Objects

1. Staging Database:

Firstly, we need to connect the SSMS to our SQL server. After that, the database AdventureWorks2014 is restored into the server through SSMS and ready for further work. After that, we have executed SQL scripts '1_AWN_STG_Demo' to construct the appropriate schema, tables, and views for the staging DB.

2. DW Database:

Then, we have executed SQL script '2_AWN_DW_Demo' (excluding procedures) to create the DW schema and tables. The procedures for populating tables were left for a subsequent stage.

Stage 2: ETL Source to Staging

In this stage we have loaded the required Data from the source AdventureWorks2014 DB to the staging AWN_STG_DEMO. The focus was on moving data from the source to the staging area using SSIS in Visual Studio.

1. SSIS Package Development:

Furthermore, for each dataflow task direct populating has been performed from source to the destination. Fig.1 displays an example of the table Currency data flow with similar configurations applied to the other tables.

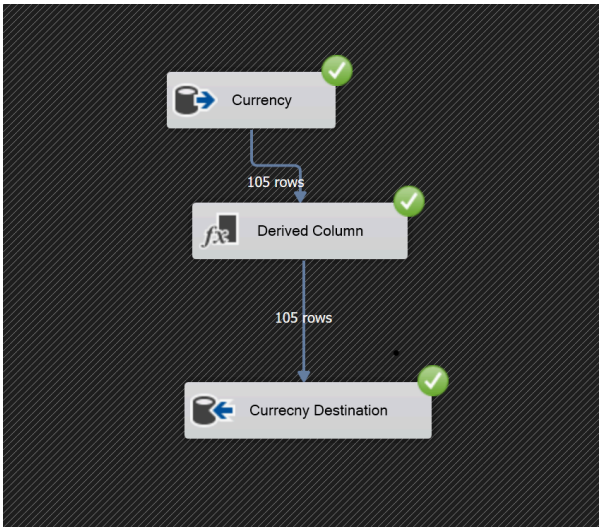


Fig. 1 Data flow

2. Verification:

Checking in SSMS for correct data population in the AWN_STG_Demo tables by excuting the command show first 1000 rows validate the population of all tables.

Stage 3: ETL Staging to DW

In this stage we have populated the dimension and fact tables of our AWN_DW_Demo DW from the staging AWN_STG_Demo DB to construct our Sale Data Mart based on the galaxy Chema provided.

First, the dimension tables have to be populated with the order provided to account for foreign key constraints. After that the fact tables can be populated.

1. Dimension Tables in DW:

Run stored procedures from '2_AWN_DW_Demo' to populate most dimension tables (except DimEmployee) and then excute each procedure manually to populate the dimensions in the DW.

2. Handling DimEmployee:

Implemented Slowly Changing Dimension methodology to maintain historical and changing attributes, Fig.2 displays the DimEmployee handling.

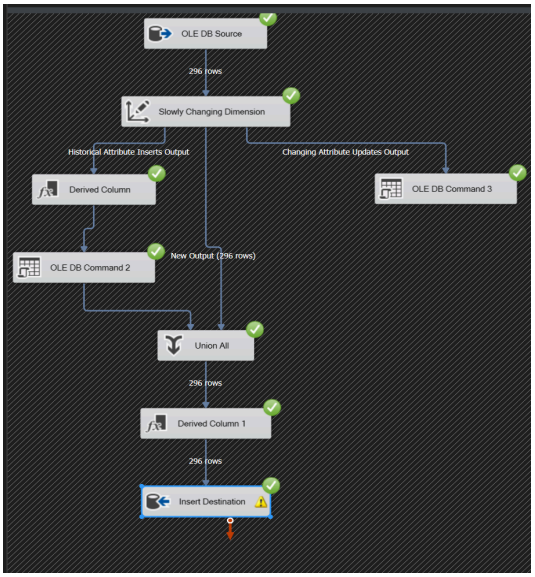


Fig.2 (Handling DimEmployee)

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3. Fact Table Population:

Finally, we have Created Data Flow Tasks to link fact tables (Internet Sales and Reseller Sales) with relevant dimensions as shown in Fig.4 (Example for the InternetSalesFact table).

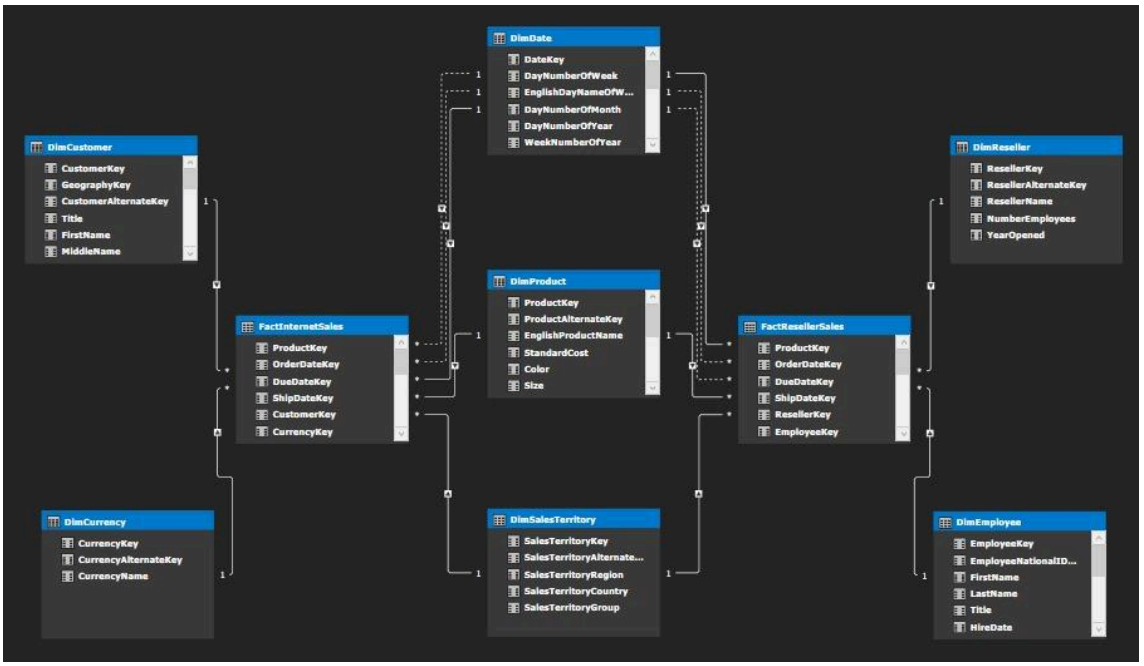


Fig.3 (InternetSales fact table example)

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Stage 4: Creating SSAS Tabular

This stage aimed to set up a tabular cube using SQL Server Analysis Services (SSAS). First step was to set up the Analysis Server and connect it to SSMS. After that, a new login authorization had to be set for the Analysis Server in order to access the database. Finally, an Analysis Service Tabular Project was created and linked to the Analysis Server.



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Stage 5: Creating Power BI Reports

Final stage is to generating and Insightful Power BI Report by linking the cube created in SSAS to the Power Bi tool.

Nevertheless, we have designed and built interactive reports showcasing various analyses of Sales across multiple dimensions. Included cumulative statistics (Total Sales, Average Monthly/Quarterly/Weekly/Daily Sales) for comprehensive data analysis. Schema of the DW can is displayed in Fig.6 generated from Power BI.

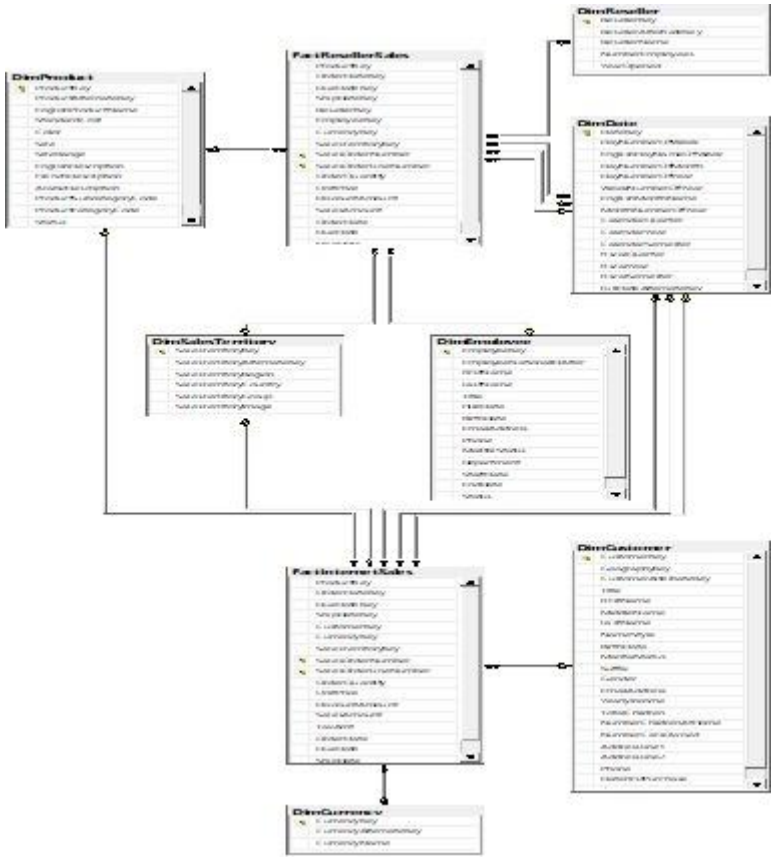


Fig.6 (DW Sales Schema)

To conclude, this report summarizes the completion of various stages in establishing the database, executing ETL processes, setting up SSAS tabular models, and generating informative Power BI reports.